

Supplementary Information

Universal Spin-Orbit Interactions in Mie Scattering

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I. Method

Electromagnetic simulations

The results of scattering patterns, scattering spectra and absorption spectra of the Au NS are obtained by solving Maxwell equations using FDTD methods. The polarization states of the diffraction orders of Au NS lattice are calculated by decomposing the electric field in a circular basis, $(\vec{e}_\theta \pm i\vec{e}_\phi)/\sqrt{2}$ in a local coordinate frame with z axis attached to the diffraction direction. The diameter of Au NS used in individual scattering and NS lattice is 80 nm and the material parameters of Au are adapted from 'Johnson and Christy'. The period of the NS lattice is 600 nm in both x and y directions. To get a more accurate picture of the evolution of the equivalent charge on the surface of Au NS, we applied the FEM method to solve Maxwell equations, which better deals with the curved surfaces. The results of equivalent charge and current, polarizability, SAM, OAM, torque and components of orbital momenta are calculated by programming the corresponding expressions in the basis of the electromagnetic fields.

Sample preparation and polarized DF characterization

The gold NSoGs and NSoMs were prepared with the method reported in (3). The gold NSs were purchased from BBI Solutions. The gold NSoGs were realized by drop-casting the NS solutions onto the glass substrates for 30 seconds, followed by rinsing with deionized water and drying with nitrogen gas. Conversely, the preparation of the gold NSoMs involved a similar procedure but preceded by the meticulous preparation of ultra-smooth gold substrates using a template stripping method. The Au substrates were immersed in a 1 mM biphenyl-4-thiol (BPT) solution for 22 h to form a 1.3 nm-thick self-assembled monolayer on the Au substrates, where the substrates were rinsed with a large amount of ethanol to remove the physically absorbed BPT molecules, and dried with nitrogen gas. Following these steps, the NSoGs and NSoMs were ready for further characterization.

The polarized DF scattering patterns were collected by a custom-built polarization-dependent single-particle DF scattering imaging system. As shown by the optical path shown in Figure 1(a), dark hollow Gaussian beams illuminated the samples from a 100 W halogen white light lamp source through a 100X DF objective (NA 0.8 Olympus LWD), and the scattering lights were collected by the same objective and imaged on a CMOS camera (Zelux® 1.6 MP Monochrome CMOS Camera). To extract the LCP and RCP components, a quarter wave-plate and a polarization beam splitter (PBS) were placed before the camera. Subsequently, the S_0 , S_3 , LCP and RCP scattering patterns from each individual nanoparticle can then be extracted from these polarized scattering patterns.

II. Supplemental figures and tables

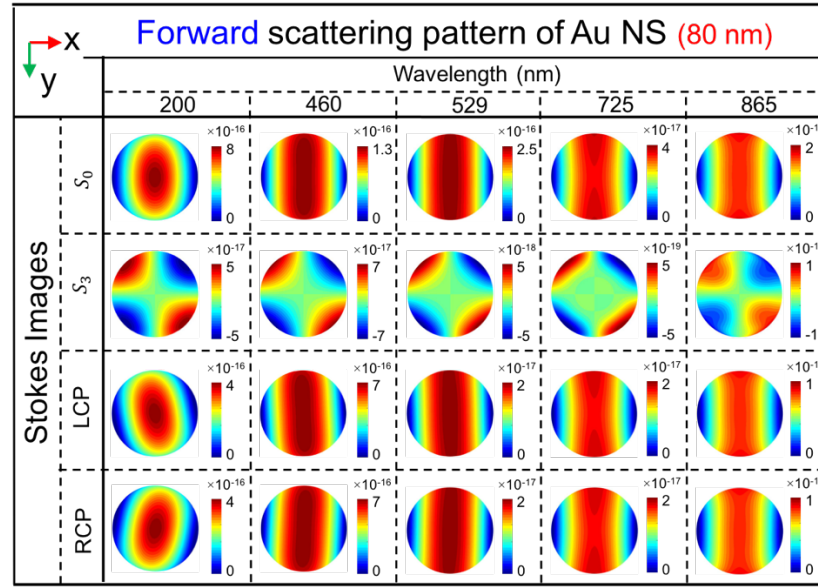


Fig. S1. The calculated forward scattering patterns of S_0 , S_3 , LCP and RCP from the Au nano-sphere at various wavelengths. The diameter is 80 nm.

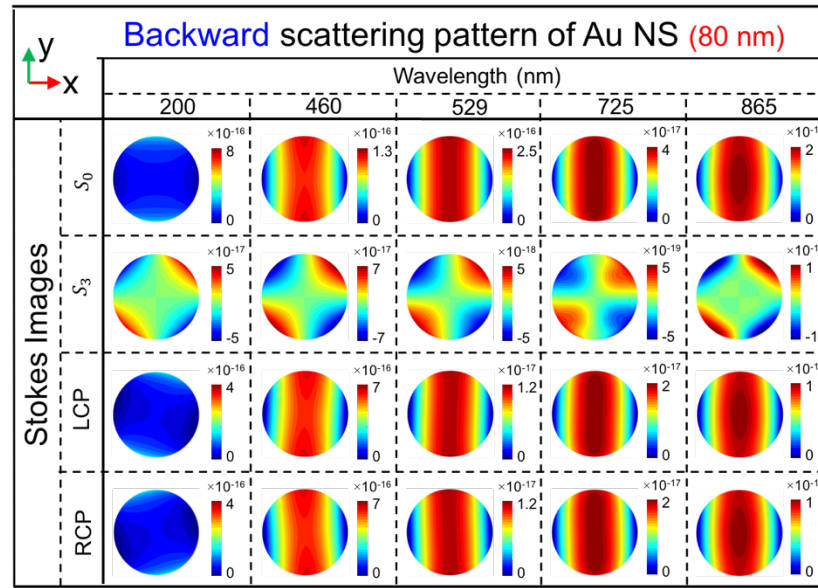


Fig. S2. The calculated forward (a) and backward scattering patterns (b) of S_0 , S_3 , LCP and RCP from the Au nano-sphere at various wavelengths. The diameter is 80 nm.

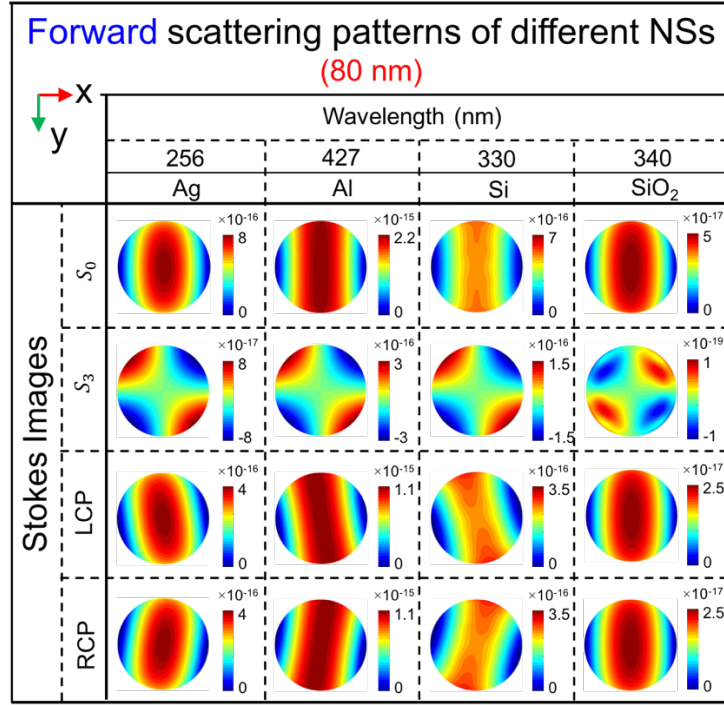


Fig. S3. Calculated forward scattering patterns of S_0 , S_3 , LCP and RCP from the Ag, Al, Si and SiO₂ nano-spheres at various wavelengths. The diameter is 80 nm. These patterns indicate that the angular shift is universal in nanoparticles.

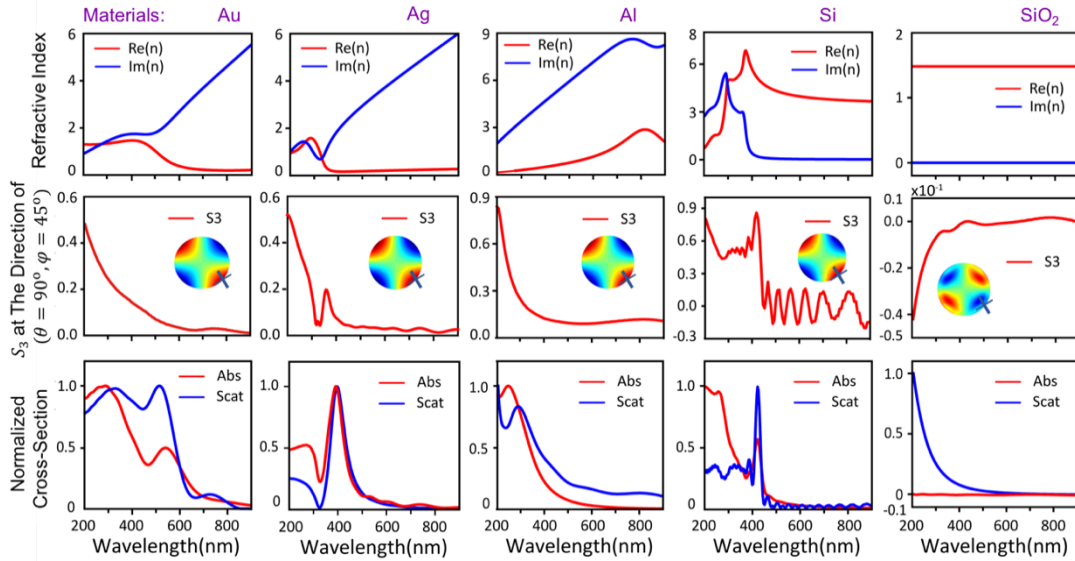


Fig. S4. (a-e) The complex refractive indices of Au, Ag, Al, Si and SiO₂. (f-j) The normalized value of S_3 in the specified scattering direction ($\theta = \pi/2$, $\phi = \pi/4$) from the corresponding nanospheres (80 nm diameter) in (a-e). The scattering direction is also marked in the inset images. (k-l) The calculated scattering and absorption cross-sections of the corresponding nanospheres. The S_3 spectra clearly do not simply depend on the resonance features of NSs, indicating that the observed angular shift is a type of non-resonant effect.

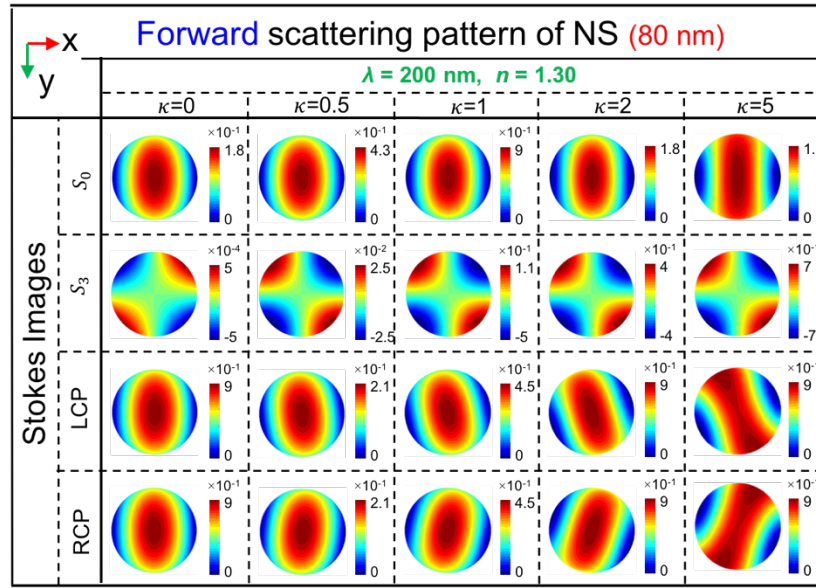


Fig. S5. The calculated forward scattering patterns from different NSs at a wavelength of 200 nm. The results are calculated by Mie theory, and the diameter of the NS is 80 nm. The real part of the refractive index is set constant to 1.3, while the imaginary part κ is varied from 0 to 5.

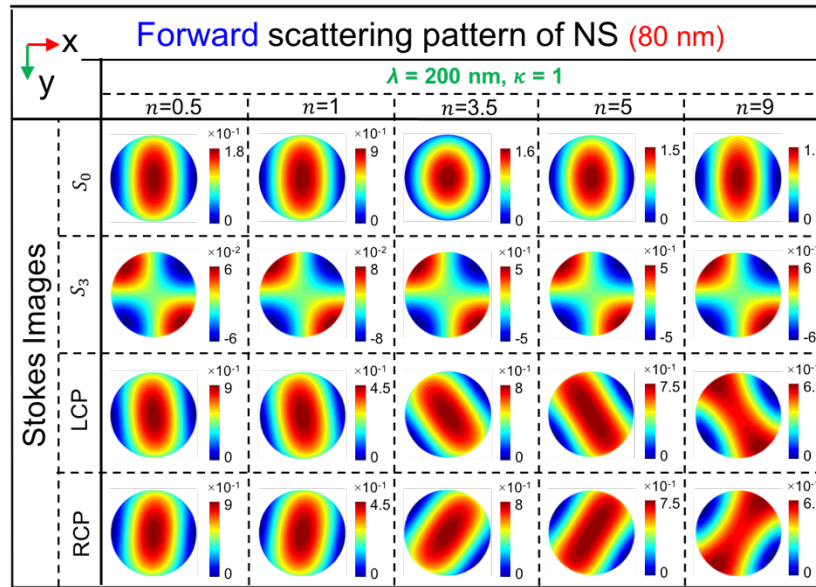


Fig. S6. The calculated forward scattering patterns from different NSs at a wavelength of 200 nm. The results are calculated by Mie theory and the diameter of the NS is 80 nm. The real part of the refractive index is varied from 0 to 9, while the imaginary part κ is set constant to 1.

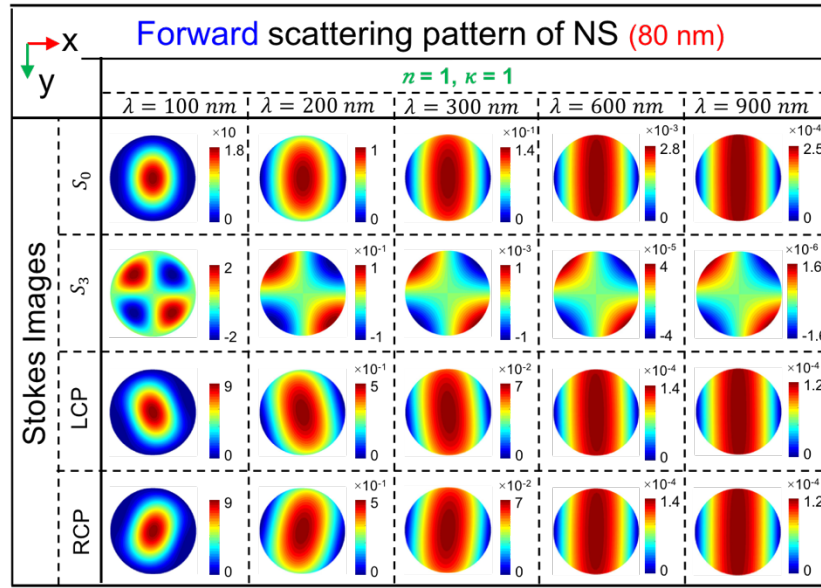


Fig. S7. The calculated forward scattering patterns from NS at different wavelengths. The results are calculated by Mie theory and the diameter of NS is 80 nm. The real part of the refractive index is set constant to 1, while the imaginary part κ is set constant to 1.

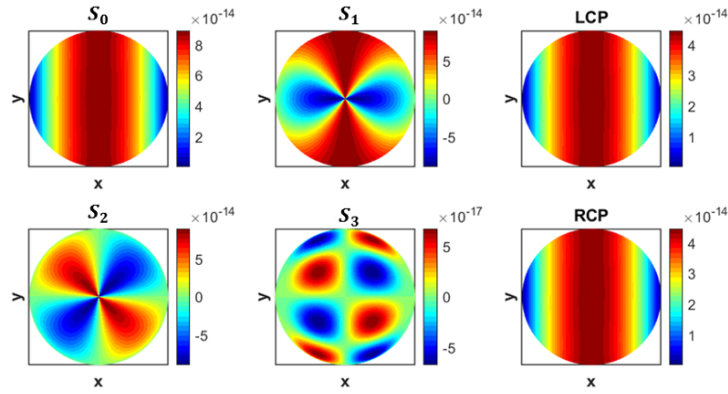


Fig. S8. The calculated forward scattering patterns of an ideal electric dipole, which oscillates along the x axis. It is clear that the LCP scattering pattern is almost the same as the RCP scattering pattern.

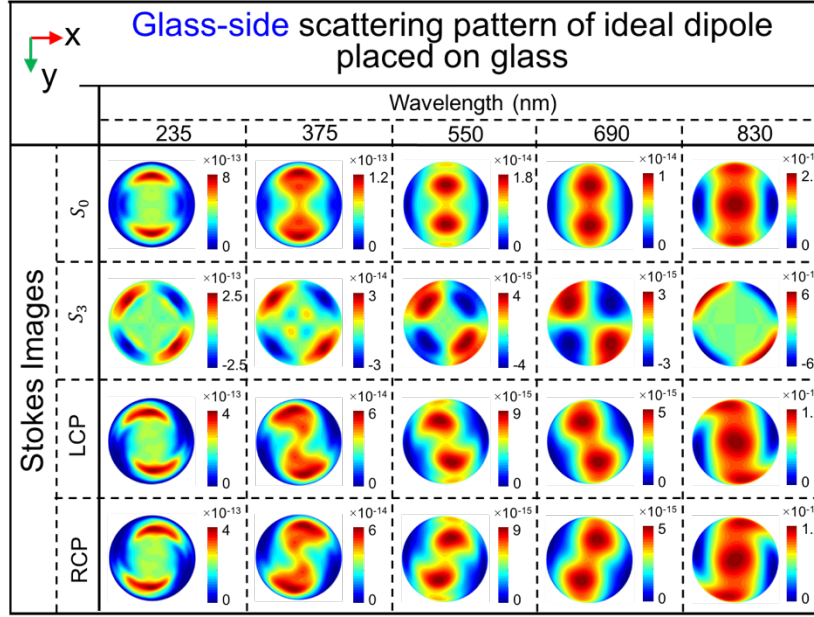


Fig. S9. The calculated forward scattering pattern of an ideal electric dipole placed on glass. The dipole is located in the air side and ~ 40 nm away from the air-glass interface. The oscillation direction is along the x axis.

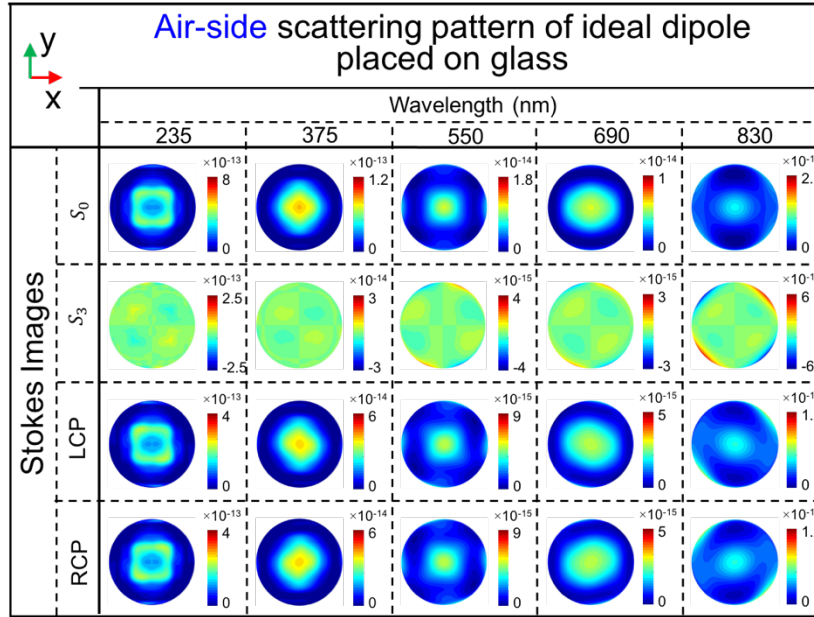


Fig. S10. The calculated backward scattering pattern of an ideal electric dipole placed on glass. The dipole is located in the air side and ~ 40 nm away from the air-glass interface. The oscillation direction is along the x axis.

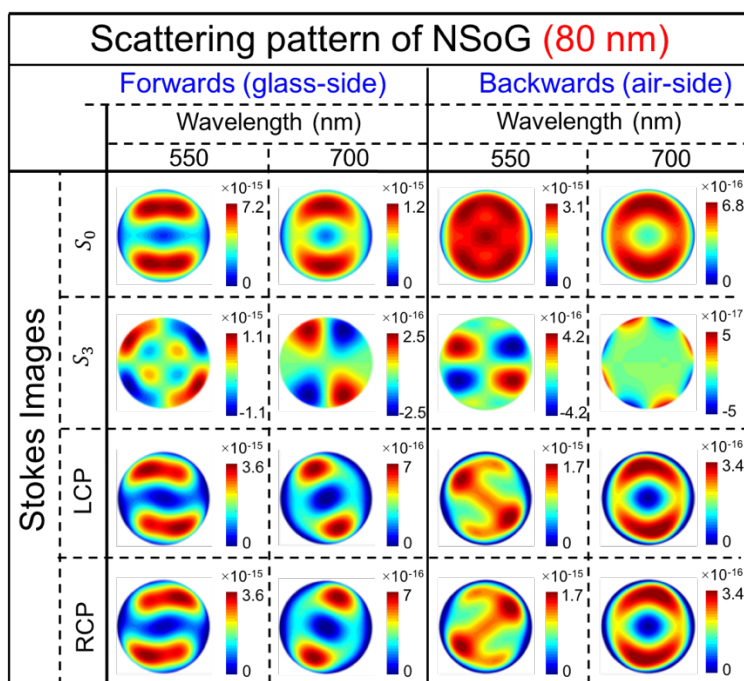


Fig. S11. The calculated scattering pattern of a Au NS placed on glass. (a) The forward scattering pattern in the glass side. (b) The backward scattering pattern in the air side. The diameter of the NS is 100 nm.

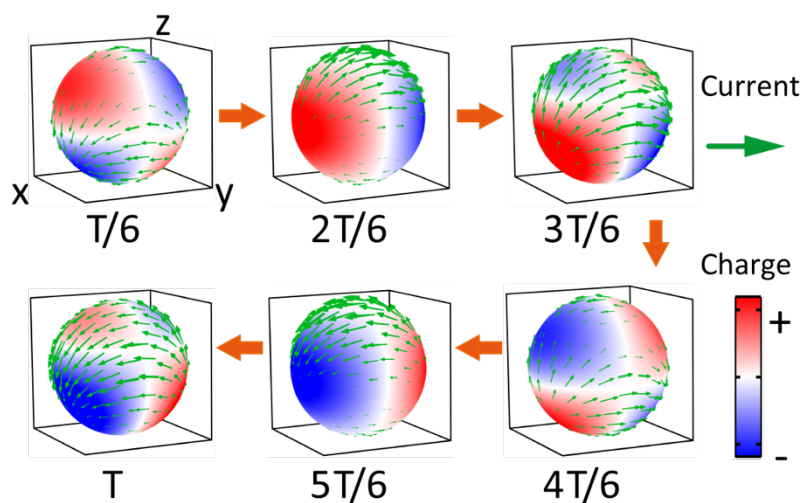


Fig. S12. The equivalent charge distribution on the NS surface and the volume current distributions for cross-section at $y=0$ and wavelength of 200 nm. The incident electric field is also displayed on the right side. The diameter of the Au NS is 80 nm. The transient distribution can be seen in **Movie S1**.

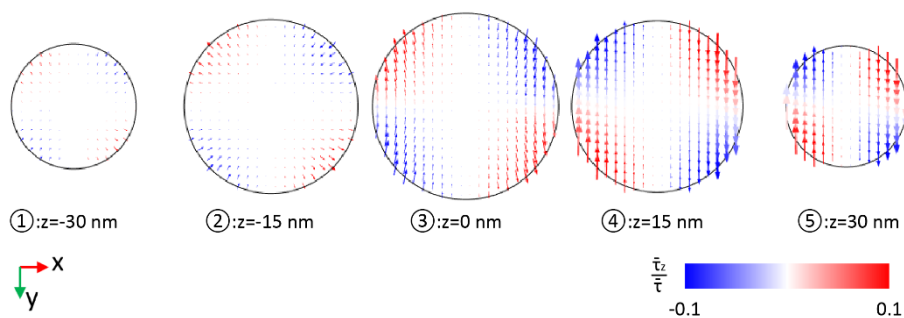


Fig. S13. The distribution of time-averaged spin torque density on x-y cross-sections of the Au NS with $z = -30$ nm, -15 nm, 0 nm, 15 nm, and 30 nm. The amplitudes of $\bar{\tau}$, $\bar{\tau}_z$ are denoted by arrow length, while the z components are given by the color of the arrow, where $\bar{\tau} = \mathbf{P} \times \mathbf{E}$. Their time evolutions are provided in **Movies S2**.

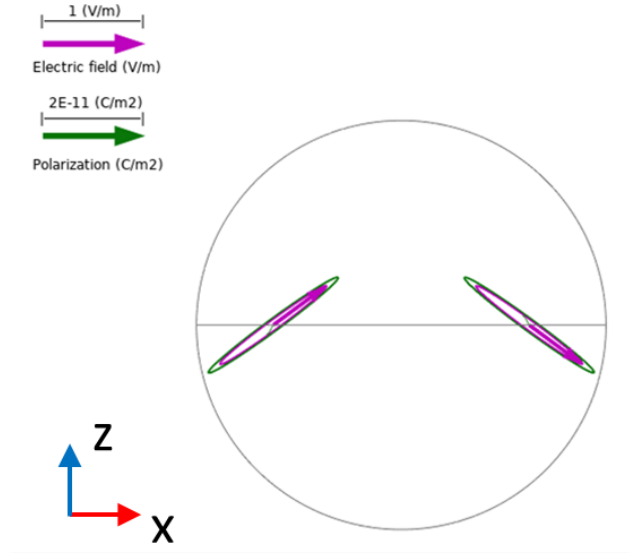


Fig. S14. The calculated rotation trajectory of the electric field (magenta arrow) and polarization vectors ($\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$, green arrow) at locations of $(\pm 25, 0, 0)$ nm. Its time evolution is provided in **Movie S3,4**.

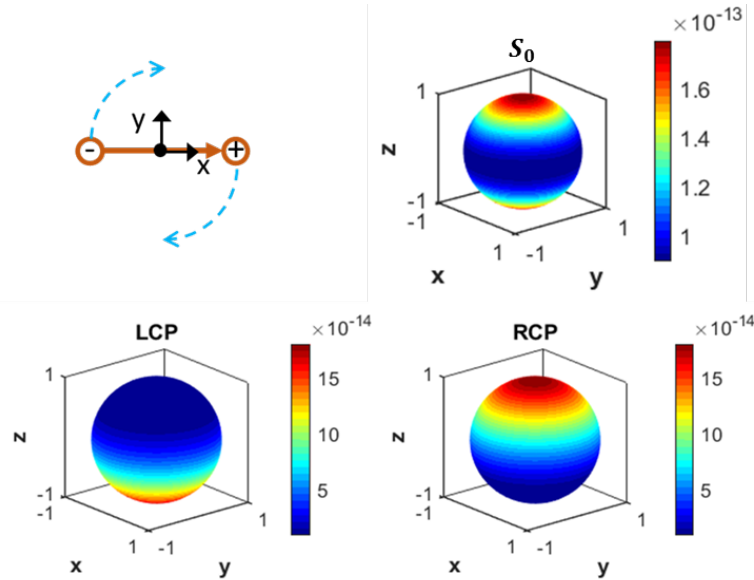


Fig. S15. The calculated scattering patterns of a spinning electric dipole, which is set as the combination of two vertical electric dipoles with 90 deg phase ahead along y axis. This spinning dipole can be realized by driving a metal NS with RCP light propagating along the $+z$ direction or LCP light propagating along the $-z$ direction. This spinning dipole emits LCP and RCP light in opposite directions, and the spin-locked emission direction depends on the rotation direction of the dipoles.

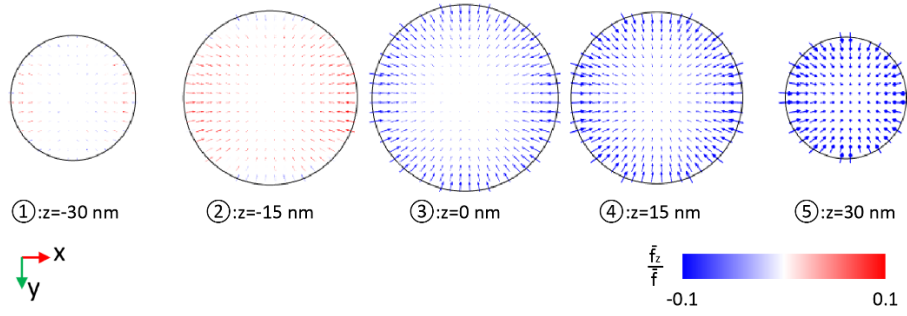


Fig. S16. The distribution of time-averaged orbital force density on x-y cross-sections of the Au NS with $z = -30$ nm, -15 nm, 0 nm, 15 nm, and 30 nm. The amplitudes of $\vec{f}$, $\vec{f}_z$ are denoted by arrow length, while the z components are given by the color of the arrow, where $\vec{f} = \mathbf{P} \cdot (\nabla) \mathbf{E} - \mathbf{E} \cdot (\nabla) \mathbf{P}$. Their time evolutions are provided in **Movies S5**.

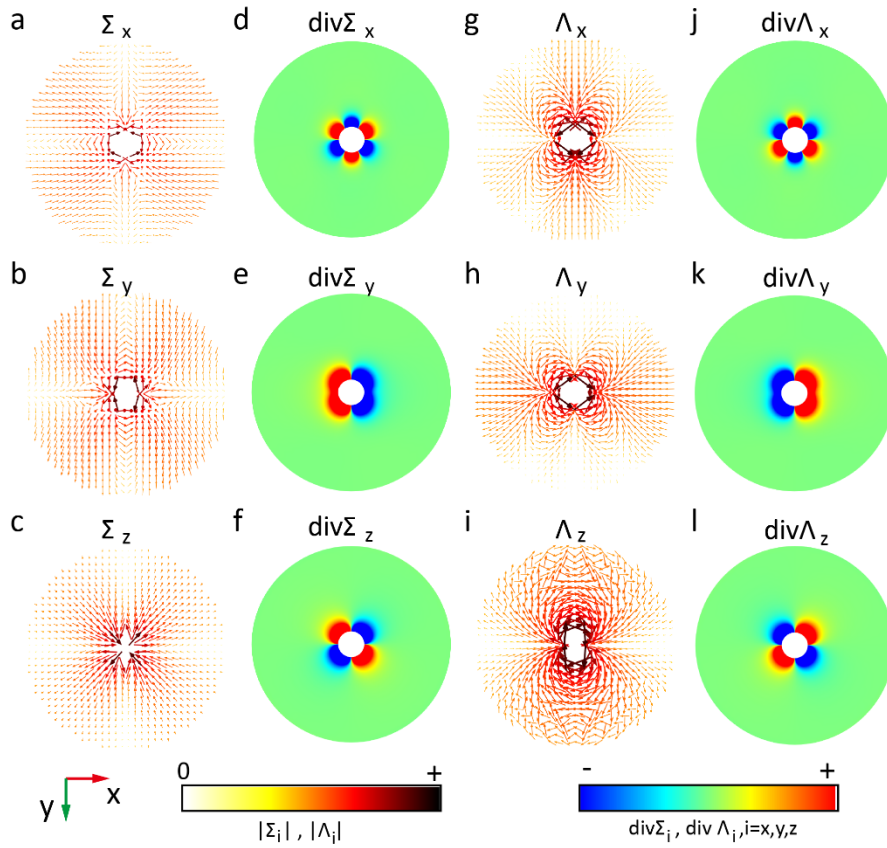


Fig. S17. The distribution of SAM, OAM flux and divergence distribution of SAM, OAM flux in the scattering field outside the NS. The time evolutions of divergence distribution of SAM, OAM flux are provided in **Movies S6**.

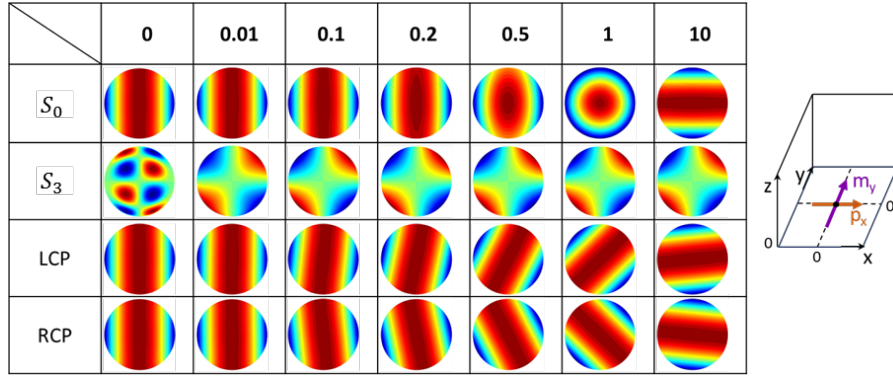


Fig. S18. The forward interference patterns between an idea electric dipole and an idea magnetic dipole. The electric dipole oscillates along x axis (p_x) and its intensity is set to 1, while the magnetic dipole is along y axis and the equivalent magnetic dipolar moment (m_y/c) varies from 0 to 10. The phase of the magnetic dipole is ahead the electric dipole of $\pi/2$.

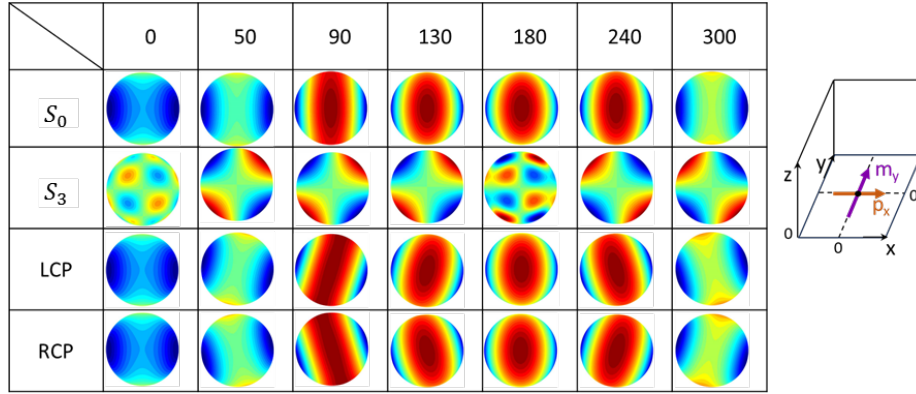


Fig. S19. The forward interference pattern between an idea electric dipole and an idea magnetic dipole. The electric dipole oscillates along x axis (p_x) and its intensity is set to 1, while the magnetic dipole is set to 0.3. The phase difference between the electric and magnetic dipoles, i.e., $\varphi(m) - \varphi(p)$, varies from 0 to 300 degrees.

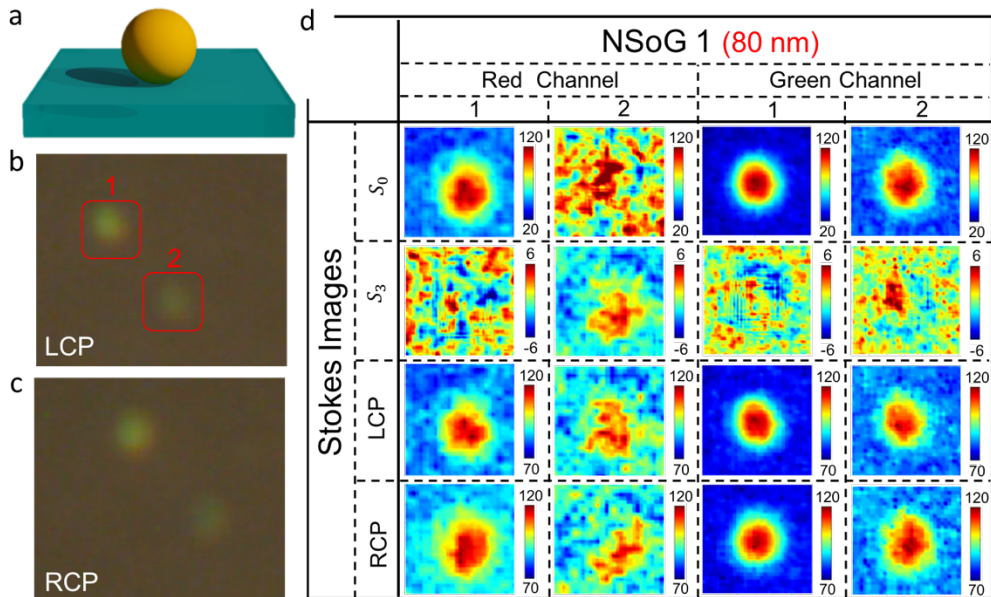


Fig. S20. Experimental verification of giant photonic SHE in nanostructures. (a) Schematic of nanosphere on glass (NSoG) with diameter of 80 nm. (b-c) The LCP (b) and RCP (c) components of the total dark-field scattering patterns. The numbers 1 and 2 denote the investigated particles. (d) The measured scattering patterns of S_0 , S_3 , LCP and RCP in the color camera red and green channels.

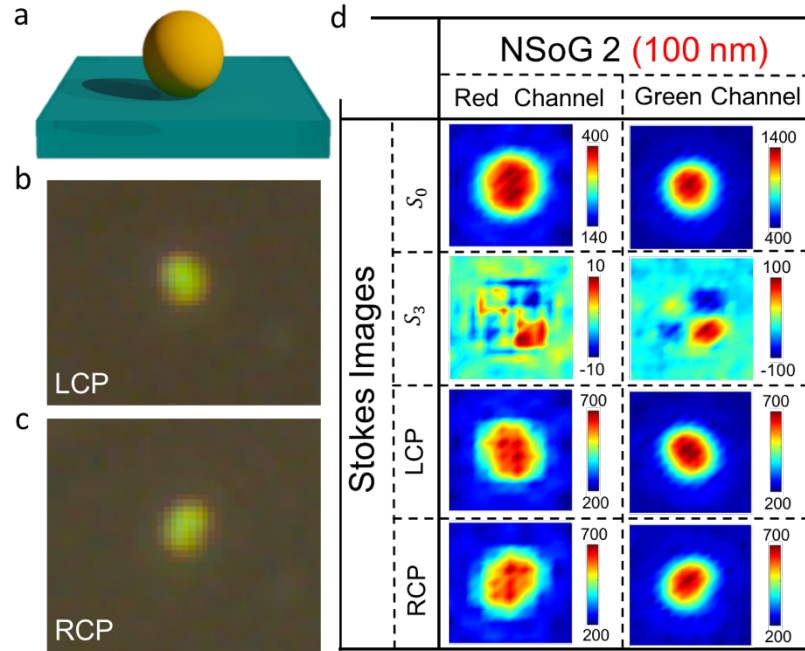


Fig. S21. Experimental verification of giant photonic SHE in nanostructures. (a) Schematic of nanosphere on glass (NSoG) with diameter of 100 nm. (b-c) The LCP (b) and RCP (c) components of the total dark-field scattering patterns. (d) The measured scattering patterns of S_0 , S_3 , LCP and RCP in the color camera red and green channels.

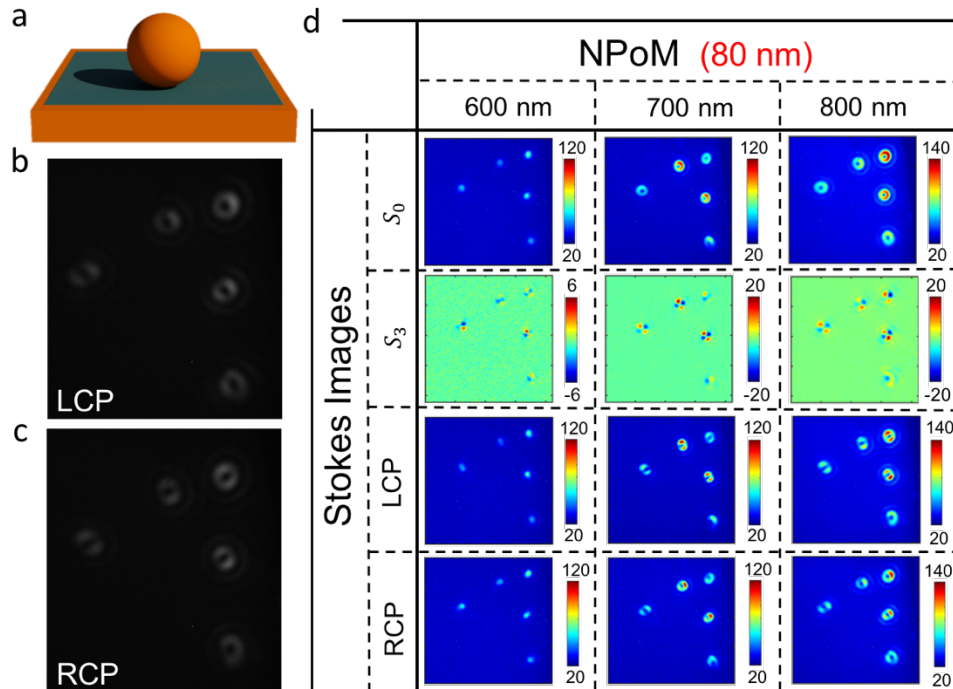


Fig. S22. Experimental verification of giant photonic SHE in nanostructures. (a) Schematic of nanosphere on mirror (NSoM) with diameter of 80 nm. (b-c) The LCP (b) and RCP (c) components of the total dark-field scattering patterns. (d) The measured scattering patterns of S_0 , S_3 , LCP and RCP at

wavelengths of 600 nm, 700 nm and 800 nm.

Reference

1. K. Y. Bliokh, J. Dressel, F. Nori, Conservation of the spin and orbital angular momenta in electromagnetism. *New Journal of Physics* **16**, 093037 (2014).
2. M. Durach, N. Noginova, Spin angular momentum transfer and plasmogalvanic phenomena. *Physical Review B* **96**, 195411 (2017).
3. S. Hu et al., Full Control of Plasmonic Nanocavities Using Gold Decahedra-on-Mirror Constructs with Monodisperse Facets. *Adv. Sci.*, 2207178 (2023).